

GRACE GATHONI KARUME

Fullstack Developer · Math & CS @ JKUAT

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SUMMARY

Software Developer specialising in React and TypeScript, with a growing full-stack foundation in Node.js and PostgreSQL. Proven ability to architect, build, and ship production-ready web applications — demonstrated through hackathon wins, a co-founded pre-seed startup, and multiple concurrent open-source and commercial contributions. Leveraging a Math & CS background to build optimised, accessible, and scalable user interfaces.

EXPERIENCE

Co-Founder & Backend Engineer · [OctalFoundry](#) March 2026 – Present

Pre-seed startup · Dev Lead & AI Pipeline Owner

- Drive all backend architecture and API design decisions as the sole technical co-founder, setting the engineering direction from day one.
- Designed and built the core backend infrastructure using Node.js, Express, and TypeScript — covering routing, auth, and business logic.
- Own the AI pipeline end-to-end: integrating, tuning, and optimising AI-powered features into the product for production use.
- Balance co-founder responsibilities — roadmap, technical hiring, and cross-functional prioritisation — alongside hands-on engineering.
- Architecting v2 as a Web3-native platform: Soulbound Tokens (SBTs) for tamper-proof proof-of-learning credentials, on-chain student data ownership via decentralised identity, and smart contract-enforced progression rules — removing the need for any centralised authority to gatekeep academic progress.

Fullstack Engineer · [MemeyAI](#) April 2026 – Present

- Designed and implemented the complete technical architecture for ArbanTV, a creator-monetisation platform enabling African creators to publish and sell short video series to paying audiences.
- Built the full frontend in React, TypeScript, and Tailwind CSS — including content discovery, series pages, and a purchase-gated video player.
- Engineered the backend with Node.js, Express, TypeScript, Supabase, and Drizzle ORM; designed the relational schema to handle creators, series, episodes, purchases, and access control.
- Delivered end-to-end type safety across the stack with shared TypeScript types, significantly reducing integration bugs at the frontend–backend boundary.

Volunteer Frontend Developer · [Buildlink](#) April 2026 – Present

- Diagnosed and resolved a broken dark mode implementation, enforcing consistent theming tokens across the entire application.
- Rebuilt and fixed a faulty login flow, improving authentication UX and substantially reducing sign-in drop-off.
- Conducted a broad UI/UX audit and shipped multiple interface improvements for better usability, visual hierarchy, and accessibility.
- Actively triaging and resolving ongoing frontend bugs as a contributing member of the core team.

Web Development Intern · [InternCareerPath \(Remote\)](#) January – March 2026

6-week intensive full-stack JavaScript programme — shipping production-ready code from week one

- Built RESTful APIs with Node.js and Express.js, implementing routing, middleware, and error handling across multiple endpoints.

- Designed normalised relational schemas in PostgreSQL; wrote efficient JOIN queries for data-heavy features.
- Integrated React frontends with Node.js backends, managing async data flow and authentication state.
- Adopted industry code-review standards and documentation practices — all work reviewed and approved by senior engineers.

PROJECTS

BuildLink Kenya — Professional Network for the Built Environment

React · TypeScript · Tailwind CSS · buildlink.co.ke

- Resolved a persistent dark mode bug where theme state failed to survive logout — fixed by correctly persisting the user preference and rehydrating it at session start, covering the sign-in/sign-up surfaces while intentionally excluding the landing page.
- Rebuilt the user profile screen: fixed colour inconsistencies in light mode, implemented a tap-to-preview popup for uploaded profile images, and resolved responsiveness issues in the portfolio upload modal.
- Shipped a richer portfolio section — added inline content editing (title, thumbnail, arrangement) and a full visual revamp with drag-and-drop organisation features.
- Improved the public profile view: surfaced the skills block as an always-present section for new users, and injected profession and company tenure into the profile banner from sign-up data.

OctalFoundry — AI-Powered Socratic Tutor for University Students

Node.js · Express · TypeScript · Supabase · Gemini API · octalfoundry.vercel.app

- Co-founded and lead engineering on a platform that transforms university coursework into career-ready skills using an AI Socratic tutor — asking probing questions rather than giving direct answers to drive genuine understanding.
- Architected the backend and AI pipeline: integrated the Gemini API to power tutoring sessions and proof-of-work verification, where the AI validates genuine concept engagement before marking progress.
- Designed an academic/holiday mode toggle that dynamically adjusts the number of assigned Foundry practice sessions, keeping students on track without burnout during breaks.

ArbanTV — Creator Monetisation Platform for African Experts

React · TypeScript · Tailwind CSS · Node.js · Express · Supabase · Drizzle ORM · arbantv.onrender.com

- Designed and implemented the full frontend architecture for a purchase-gated video platform — creators publish short expert series; viewers unlock access per purchase.
- Delivered the complete frontend from concept to deployment: content discovery, series pages, episode viewer, and purchase flow, all in React, TypeScript, and Tailwind CSS.
- Contributed to system design across the stack and currently building out the backend — relational schema, auth, and payments — using Node.js, Supabase, and Drizzle ORM.

TECHNICAL SKILLS

Frontend: React · TypeScript · JavaScript (ES6+) · HTML5 · CSS3 · Tailwind CSS

Backend: Node.js · Express.js · PostgreSQL · RESTful API Design · Drizzle ORM

AI & Data: AI pipeline integration · Chart.js · Leaflet.js · OpenStreetMap API

UI Libraries: ShadCN UI · CSS Modules

Tools: Git / GitHub · Figma · Linux · Supabase · Postman · Third-party API Integration

CS Fundamentals: Data Structures · Algorithms · OOP · Discrete Mathematics · Responsive Design

EDUCATION

BSc Mathematics & Computer Science*Expected 2027*

Jomo Kenyatta University of Agriculture and Technology (JKUAT)

Coursework: Data Structures · Algorithms · Object-Oriented Programming · Discrete Mathematics